

CNC ROUTER

Computer-Controlled Routing of Sheet Goods — anatomy · motion & electronics · tooling · feeds & speeds · workflow · G-code · safety

24,000 rpm	spindle speedsinto the top of the range
2000x4000 mm	travel on largeflat-bed machines
1–10 m/min	typical feed rate —mmmoves fast over sheet
±0.1–0.3 mm	achievableinpositioning tolerance
3–5 axes	XYZZ plus rotaryA and tilting C axes

THE BASICS

WHAT IS A CNC ROUTER?

A CNC (Computer Numerical Control) **router** is a computer-driven machine that spins a routing cutter along **X** (left–right), **Y** (front–back) and **Z** (up–down) to cut parts from **wood**, **MDF**, **plywood**, **plastics**, **foam** and **soft aluminium**. A person designs the part on a computer; motors follow programmed toolpaths with ±0.1 mm precision — far beyond hand routing. It is the subtractive, non-metal-biased cousin of the mill: the mill cuts hard metal block with a rigid column and high torque, while the router trades stiffness for a huge flat **table area** and high spindle speed — built to part sheet goods fast.

Key ideas

- Subtractive** — starts from a solid sheet/block and removes material (the opposite of 3D printing)
- 2D profiles**, **2.5D pockets**, **full 3D carved shapes** — flat bars, routed letters, sculpted reliefs
- Rotary axes** (A = rotate, C = tilt) make 4- and 5-axis machines for cylinders and undercuts
- G-code** drives every move: a numbered programme of positions, speeds and feeds
- History**: NC demoed in 1952 (MIT + Parsons); hobby CNC exploded ~2010 with Arduino + GRBL

What it is **not**: not a 3D printer (those add layers), not a laser (no smoke or kerf — but no sharp zero-radius corners either), and not a mill (no steel hogging). For flat sheets and prismatic parts it is the fastest path from a CAD file to a finished part on a hobby budget.

MACHINE ANATOMY

ANATOMY OF A ROUTER

Part	Role
Frame	Rigid steel/aluminium base or gantry — rigidity = accuracy
Spindle / router	Holds & spins the bit (trim router or VFD spindle)
Linear rails	Guides each axis — HGR rails, V-wheels, round shafts
Ball screw / rack	Converts motor rotation into straight motion
Stepper / servo motors	Move the axes (NEMA 17/23/34), closed-loop servos
Controller	Runs G-code and drives the motors (GRBL, FluidNC, Mach3...)
Vacuum table	Holds sheet stock flat by suction through a spoilboard
Dust collection	Chip + dust extraction — dust shoe, hose, cyclone, filter

Two frame dialects: **gantry** (bridge rides over a stationary table — big, open, holds stiff sheet) and **moving-table** (table feeds into a fixed bridge — compact, heavier work). **Spindle/router** motor supplies the rotation; **ball screws/linear rails** give smooth, accurate travel; the **controller** is the brain that turns G-code lines into electrical pulses.

WHY THEY WIN

10 ADVANTAGES OF CNC ROUTING

- Precision & repeatability** — ±0.1–0.3 mm, identical part after part, no human fatigue
- Complex 2.5D/3D carving** — sweeping pockets, inlays, sculpted reliefs a hand router can't cut
- Automation** — load a sheet, press start, walk away; run overnight for production
- Low waste** — nested CAD parts pack tightly into a sheet, cutting scrap dramatically
- Small-radius interior features** — sharp inside corners limited only by bit radius (~1.5 mm)
- Fast on sheet goods** — 1–10 m/min feeds move across big panels quickly
- Full 3D** — ball-nose bits machine smooth curved reliefs and moulds
- Repeatable joinery** — mortise & tenon, finger joints cut to snap-fit accuracy every time
- Scalable** — hobby desktop to 2x4 m production flatbeds
- Software driving hardware** — CAD/CAM changes a design and re-cuts in minutes

CIRCUITS & DRIVES

MOTION SYSTEMS

The accuracy of a router lives or dies in its **drive train**: how motor rotation becomes tool movement, and how rigid the whole loop is.

Drive	Speed	Accuracy	Cost	Notes
Lead screw (ACME + antibacklash nut)	0.5–2 m/min	0.1–0.3 mm/300	\$	simple, cheap; nut wears
Ball screw	to 10 m/min	0.02–0.05 mm/300	\$\$\$	recirculating balls, near-zero backlash — pro standard
Rack & pinion	10 m/min+	0.1–0.5 mm/300	\$\$	long beds; spring-loaded antibacklash pinion
Belt drive	fast	medium	\$	stretch + backlash; light duty

Stepper vs servo: steppers (NEMA 17/23/34) are open-loop — cheap and strong but can **skip steps** when overloaded; servos carry **encoders** that close the loop, so no lost steps, plus more speed and torque for a higher price. **Rigidity matters**: a flimsy frame or springy linear rail ruins even a perfect ballscrew — stiffness is what converts motor force into a clean cut.

Accuracy implication: rack-and-pinion wins on long beds where a long ballscrew would whip under load; ball screws win where repeatability matters most. The longest axis (usually X) is where these trade-offs bite hardest.

CUTTING POWER

SPINDLES & MOTORS

	Trim router	VFD spindle
Example	Makita RT0701, DeWalt DW611	1.5–3 kW, 3-phase + VFD
Speed	10,000–30,000 rpm	6,000–24,000 rpm
Torque	little at low rpm	constant torque
Cooling	air (fan)	water- or air-cooled
Collets	1/8–1/2 in.	ER11 / ER20 / ER25 / ER32
Noise	LOUD (100 dB+)	quiet
Runout	0.01–0.05 mm	under 0.01 mm
Cost	~US\$80	~10x the cost

Tool holding is via **ER collets** — spring collets that squeeze the bit shank: **ER11 = 1–7 mm** · **ER20 = 1–13 mm** · **ER25 = 1–16 mm** · **ER32 = 2–20 mm**. **kW/hr rating** states continuous cutting power and how long the motor holds full torque. Seat the bit fully, keep collets clean, and never run with a loose nut — runout (wobble) above ~0.05 mm means rough cuts and snapped bits.

The **hours** part of the rating says how long the motor sustains full power before thermal cut-out; a 3 kW spindle with a good rating carves hardwood all day without sagging. **Air-cooled** spindles are simpler but noisier; water-cooled are quieter and hold temperature steadier on long runs.

END MILLS

BITS & TOOLING

Bit	Best for	Notes
Up-cut spiral	fast chip removal	rough bottom, tearout on top; pairs with dust
Down-cut spiral	clean top edge	pushes part down; chips pack in slot
Compression	plywood, sheet goods	up-cut tip + down-cut top = clean both sides
Straight / flat flute	pockets & profiles	square corners; 2- or 4-flute
Ball nose	3D carving	finish passes at small stepover
V-bit (60°/90°)	engraving, signs	v-carved letters and chamfers
Engraving / drag	fine text & detail	light cuts, sharp carbide
O-flute (single)	acrylic, plastics	one big flute clears chips, no melting

Direction: the helix pulls or pushes chips — up-cut lifts them out (fast but rough top), down-cut presses the part down (clean top), compression does both for laminates. **Shank** (3.175/6.35 mm metric, 1/8–1/2 in.) is gripped by the collet; **cutting diameter D** sets slot width and stepover; inside corners can **never be sharper than the bit radius**. Shortest bit that reaches always — deflection grows with length cubed.

THE CUTTING FORMULA

FEEDS & SPEEDS

Feed (mm/min) = Chipload (mm/tooth) × RPM × number of flutes — chipload is the thickness of the chip each tooth peels off. Worked example: a 6 mm 2-flute bit in softwood at 16,000 rpm with chipload 0.15 → 0.15 × 16,000 × 2 = 4,800 mm/min. Start slower and tune by ear.

Material	Bit	RPM	Chipload mm/flute	Tips
Softwood	2-flute	16–20k	0.10–0.25	fast; watch grain tearout
Hardwood	2-flute	14–18k	0.07–0.15	shallow passes, sharp carbide
Plywood	compression	14–18k	0.08–0.20	compression kills fuzz
MDF	2-flute	14–18k	0.08–0.18	very dusty — extract!
Acrylic	O-flute	12–16k	0.05–0.13	low feed = melting
Aluminium	2–3 flute	8–16k	0.02–0.06	thin cuts, lube, climb
Foam (XPS)	2-flute	10–18k	0.25–0.50	huge chiploads fine

Depth per pass: max 1× bit diameter softwood, 0.5× hardwood, 0.2–0.4× aluminium. **Stepover**: 40–50% diameter roughing, 5–10% finishing. **Climb vs conventional**: climb (cutter rotates with the feed) gives a clean finish but pulls the cutter in — needs a rigid anti-backlash machine; conventional pushes the part away, safer on loose machines. **Plunge rate**: ramps and helical entries beat straight plunges.

Plunge tips: ramp in or use **helical entry arcs** instead of straight plunges — they keep chip load even and avoid chatter at the cut start. **Chipload is the single most useful number to tune**: too low means the bit rubs and burns (raise feed or lower rpm); too high means bogging and breakage (reverse). For any new material, start at the low end of the range with a shallow pass, listen to the whine, then push the numbers up on scrap.

CONTROL & CAM

SOFTWARE & G-CODE

Two software halves: CAM decides how to cut, controller drives the machine.

Category	Tools
Controller	GRBL (Arduino), FluidNC (ESP32), Mach3/4, LinuxCNC, Estlcam
CAM	VCarve, Fusion 360 CAM, Easel, Carbide Create, Carveco
G-code sender	Easel, UGS, Candle, bCNC, gSender

G-code basics

- G00** rapid move · **G01** straight cut at feed
- G02** / **G03** arc CW / CCW · **G17** XY plane
- G20** / **G21** inches / millimetres · **G90**/**G1** absolute / incremental
- G28** return to machine home · **M3**/**M5** spindle on/off
- Sample: **G21 G90 ; G0 X0 Y0 Z5 ; M3 S16000 ; G1 Z-3 F300 ; G1 X60 Y40 F1500 ; G0 Z10 ; M5 M30**

KNOW THE LIMITS

LIMITATIONS

- Sheet / flexible-ish materials only** — built for panels, not solid blocks taller than reachable Z
- Softer than a mill** — the gantry flexes, so it can't hog out steel or hard alloys
- Slower metal removal** — for aluminium expect light passes at modest depth, not heavy hogging
- Dust & noise** — fine wood dust is a health hazard; spindles push past 100 dB (extraction + hearing required)
- Not for hard metals** — steel, titanium and hardened alloys need the rigidity of a mill
- Bit-size limits** — inside corners always carry the bit's radius; very fine detail needs tiny fragile cutters
- Flat work bias** — prismatic sheet parts shine; thick, three-dimensional billets don't
- Setup & safety burden** — workholding, touch-off and feeds tuning take skill to do well

Choose a router when the work is **sheet-sized** and the material is wood, plastic, foam or soft aluminium. The instant a job demands hard metal, tight prismatic tolerances, or a thick 3D billet, step up to a mill.

STAY SAFE

SAFETY FIRST

- Dust mask + extraction** — wood dust harms lungs; keep a cyclone + filter running, never let chips pile up
- Eye & ear protection** — spinning cutters and 100 dB routers demand both, every time
- No loose clothing, hair, gloves or jewellery** — anything dangling gets dragged into the spindle
- Spindle guards & enclosures** — shield the spinning tool; a **dust shoe** doubles as a guard
- Emergency stop** (E-stop) within reach — test it every session; never leave a job unattended
- Secure stock** — a loose part becomes a projectile; verify hold-down before every run
- No reaching in while the spindle spins** — wait for a full stop
- Tighten collets & check runout** each job; new bit = air test first
- Fire risk** — burning wood or smoke means feeds/speeds are wrong; fix before continuing
- Unplug to change bits**; keep a fire extinguisher and first-aid nearby

COMPARISON

CNC ROUTER vs CNC MILL vs LASER CUTTER

The following table is not by means of quality and cost of the machine.

	CNC router	CNC mill	Laser cutter
Method	Subtractive — spinning routing bit	Subtractive — rigid rotating end mill	Burn / vaporise with focused beam
Core strength	Huge flat table, fast sheet work	Rigid column, high torque on metals	Ultra-fine kerf (0.1–0.3 mm)
Materials	Wood, MDF, ply, plastics, foam, soft Al	Steel, aluminium, titanium, brass	Wood, acrylic, leather, card (not most metals)
Accuracy	±0.1–0.3 mm	±0.005–0.02 mm	±0.01–0.1 mm
3D / thickness	2D, 2.5D, full 3D on sheet	Full 3D on solid billets	Flat thin stock only (2D / engrave)
Detail limit	Bit diameter (~0.8 mm min)	Minute end mills, tight corners	Very fine; no physical tool

VOCABULARY

GLOSSARY — 11 KEY TERMS

Spindle — motor + collet that spins the bit

G-code — move programme the machine executes

CAM — software that turns a model into toolpaths

Toolpath — the path the bit follows through the stock

Stepover — sideways distance between passes (% of bit diameter)

Chip load — material removed per tooth per rev (mm/tooth)

Climb milling — cutter rotates WITH the feed — clean finish, needs rigidity

Spoilboard — replaceable MDF wasteboard you cut into; resurface to keep flat

Vacuum table — table that holds sheet stock by suction through the spoilboard

Collet — spring chuck (ER-11/16/20/32) that grips the bit shank

Plunge bit — any router bit that feeds vertically straight into the stock, often a spiral end mill with end-cutting teeth